

link), automotive, broadcasting video, industrial automation, printers and medical imaging applications.

Serial standards and protocols like PCI Express, serial rapid I/O and SerialLite are accepted in high-speed serial transceivers. "These high-speed serial transceivers supporting speed of up to 10.3125 Gbps (also called 'multi-gigabit transceivers') are now integrated in high-end FPGAs offering the benefits of less board space and increased flexibility and eliminating the interfacing issues for the board designers," shares Varma. High-speed serial solutions are applicable in next-generation telecommunication, networking and storage applications.

Fusion mixed-signal FPGAs

There is a common understanding that FPGAs are only digital devices. However, there are certain mixed-signal field-programmable gate arrays that integrate programmable analogue fabric and digital fabric. Such mixed-signal FPGAs find applications in power management, smart battery charging, clock generation and management, and motor control, which are still served by discrete analogue components, mixed-signal ASICs or microcontroller units with embedded analogue-to-digital converters (ADCs). An MCU with embedded ADC has limitation on the number of analogue inputs and the speed at which it oper-



ates is relatively low.

Analogue control can be differentiated from digital control by a simple example of the dimming light application. Analogue signals process the output continuously as opposed to digital control of the dimming in steps. In short, real-world applications are more associated with the analogue domain, while digital comes in when you need to store and process the data.

The Fusion family from Actel is claimed to be the first embodiment of mixed-signal FPGAs. The architecture supports a 12-bit internal ADC at a sampling speed of 600 kSa/s, eliminating the need for external mixed-signal support ICs. It can take 30 high-voltage-tolerant analogue inputs. Fusion devices eliminate the need for multi-voltage and power-up sequencers. These operate off a single 3.3V supply. The combination of single-voltage operation and power management makes them ideal for system/board

management applications.

FPGAs with embedded processor cores

Inclusion of memory and other blocks on FPGAs have made them suitable for implementing the entire system-on-a-chip with in-built processor core. The advantage of implementing processor and its mix in the programmable logic fabric is the immense hardware-software flexibility. There are two approaches provided by major FPGA vendors for the embedded processor core. The hard-core processor has dedicated silicon on the FPGA chip, while you can implement soft-core entirely using the hardware descriptive language (HDL). As hard core is fixed on the chip, it does not provide the flexibility of soft cores.

The first processor core designed by ARM along with Actel specifically for implementation in FPGAs is 32-bit Cortex M1. The ARM Cortex M1 de-

Selection criteria for FPGAs for your application

(Based on collective opinion of contributors)

- Number of logic cells
- In-built memory requirement
- In-built processor
- Number of inputs/outputs required
- Does your application need high performance and repetitive DSP
- IP cores needed for your application
- Tools and support from the vendor
- Expertise and experience of senior engineers in your organisation
- Power requirement of your application
- Cost